Asymptotic i.i.d. prophet inequalities: fixed-capacity limits and certified two-unit bounds

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Abstract

For every fixed positive integer k, we prove existence of the worst-case asymptotic ratio between the optimal online reward from at most k selections among n i.i.d. nonnegative observations and the expected sum of the k largest observations. The observation law is known to the decision maker and may vary with n. The limit equals a continuum minimax value with prophet profile $P_k(x) = \mathbb{E}\min\{\mathrm{Pois}(x), k\}$. The proof combines compactness of cumulative acceptance functions, approximation at both quantile boundaries, and quantitative finite-horizon transfers in both directions. For $k = 2$, explicit ball-arithmetic certificates give $0.83695 \le \gamma_2^* \le 0.8372914$ and strictly separate this limit from a previously proposed nonlinear-system value. The lower witness is a mixture of 36 quantile policies; the upper witness is a measure with 41 layer atoms, not an observation law. An occupancy-weighted construction, consistent with earlier finite-horizon flow formulations, shows that finite mixtures can be dominated by deterministic time- and state-dependent quantile thresholds. The numerical certificates are specific to $k = 2$; no equality with the all-horizon infimum or attainment by a bounded threshold policy is claimed.

1 Introduction

Let $X_1, \dots, X_n$ be independent with common known law F on $[0, \infty)$ and $0 < \mathbb{E}X_1 < \infty$. A decision maker sees the observations sequentially and may irrevocably accept at most k of them. For $n \ge k$, let $\mathrm{DP}_{n,k}(F)$ be the optimal expected accepted sum, and let $\mathrm{PR}_{n,k}(F)$ be the expected sum of the k largest observations. We study

$$\gamma_{n,k} = \inf_F \frac{\mathrm{DP}_{n,k}(F)}{\mathrm{PR}_{n,k}(F)}.$$

The infimum permits a different law at each horizon. Consequently, existence of a limit for each fixed law does not establish convergence of $\gamma_{n,k}$.

The main analytic result, Theorem 2, proves convergence for every fixed k and identifies the limit with a continuum game. The compact strategy variable is the cumulative acceptance function, not a bounded set of thresholds or a compact family of observation laws. Lower transfer is uniform over laws for each fixed finite policy mixture. Upper transfer constructs a horizon-dependent finite-support law from each fixed step profile. Compactness and explicit origin and tail estimates connect these two directions. These arguments do not interchange a limit with an infimum over all laws.

Relation to earlier formulations. For $k = 1$, the asymptotic constant and its optimal-policy interpretation are classical [7, 8, 4]. Jiang, Ma and Zhang [9, Definition 4.1, Eq. (25), Theorem 4.2] give a finite-horizon semi-infinite linear programme for arbitrary k, n using state occupancy and cumulative type coverage; their Proposition 4.8 controls quantile discretization at fixed n. Our cumulative acceptance formulation is closely related to that framework. Their Lemma 2.11 and Appendix C also supply the relevant threshold-recovery precedent. We record an explicit continuous-time occupancy formula, not a new purification principle.

Brustle, Perez-Salazar and Verdugo [1, Theorems 1–2] give an exact finite-horizon quantile formulation and an asymptotic lower bound

$$\gamma_{n,k} \geq \left(1 - \frac{24k \log^2 n}{n}\right) \sum_{j=1}^{k} \theta_j^\star$$

for sufficiently large n. This is a lower bound on a liminf, not an identification of the optimal limit. Molina, Gast, Loiseau and Perchet [16, Proposition 5.4, Eq. (5.7)] analyze a related boundary-value construction; with equal selection and benchmark budgets it yields the same kind of lower bound. Their fixed-horizon finite-support reduction (Proposition 5.5) is distinct from convergence as the horizon grows. The asymptotic tightness question is also discussed in [2].

Quantile-kernel minimax and finite-horizon implementation are substantive methodological precedents in Loiseau et al. [15, Section 3, Appendix C.4]. Their principal model uses a single static threshold and a different endpoint topology. Here the controls depend on time and remaining budget, the benchmark is the top-k sum, and compactness is imposed on cumulative coverage; both endpoint estimates are proved explicitly. Other quantile and restricted-policy formulations include [12, 6], and related online-selection analyses include [13]. Kong and Murthy [11] study fixed-law extreme-value asymptotics, which do not justify moving the infimum over laws through the limit. Cruz-Ossa, Perez-Salazar and Verdugo [3] study single-threshold competition complexity with a potentially different observation budget. Neither comparison supplies the unrestricted adaptive limit proved here. A bounded source comparison found no earlier theorem with precisely this fixed-budget worst-case limit and characterization; this is not an exhaustive priority claim.

Certified two-unit consequence. The analytic theorem also makes finite continuum certificates sufficient for asymptotic bounds. The following numerical consequence uses the explicit witnesses and rigorous arithmetic described in Section 3.

Theorem 1 (computer-assisted numerical bounds)**.** *The limit* $\gamma_2^* = \lim_{n\to\infty} \gamma_{n,2}$ *exists, and*

$$0.83695 \leq \gamma_2^* \leq 0.8372914 < 0.83730.$$

Moreover

$$\theta_1^\star + \theta_2^\star < \frac{5723787}{6850000} < 0.835590, \qquad \gamma_2^* - (\theta_1^\star + \theta_2^\star) > \frac{18641}{13700000}.$$

Thus the nonlinear-system value is not the asymptotic optimum for two units. This does not settle a claim about $\inf_n \gamma_{n,2}$, or show where a particular restricted dual construction loses value. The strengthened certified bracket has width 0.0003414; rounding the upper endpoint outward to 0.83730 gives width 0.00035. Numerical single-horizon bounds in [9, 1] and fixed-law numerical asymptotics in [11] are useful comparisons, but are not substitutes for this asymptotic certificate.

2 A continuum characterization for every fixed number of selections

Fix an integer $k \geq 1$. For $n \geq k$, let $\gamma_{n,k}$ be the infimum, over laws F on $[0,\infty)$ with $0 < \mathbb{E}_F X < \infty$, of the optimal expected reward from at most k online selections divided by the expected sum of the k largest of n independent observations. The restriction $n \geq k$ is immaterial for the limit; for smaller horizons one may define the benchmark using the largest $\min(k,n)$ observations. Set

$$P_k(x) = \mathbb{E}\min\{\mathrm{Pois}(x), k\} = \int_0^x \Pr(\mathrm{Pois}(s) \leq k-1)\, ds.$$

Then $P_k(0) = 0$, $P_k'(0) = 1$, $P_k(\infty) = k$, and

$$P_k''(x) = -e^{-x}\frac{x^{k-1}}{(k-1)!} \leq 0, \qquad 0 < P_k(x) \leq \min\{x, k\} \quad (x > 0).$$

In particular $P_k(x)/x \to 1$ as $x \downarrow 0$, including when $k = 1$.

A nonnegative bounded measurable threshold vector $a = (a_1, \ldots, a_k)$ accepts an arrival of scaled tail quantile s at time t with j slots remaining when $s \leq a_j(t)$. Use column state probabilities indexed by $0, \ldots, k$, initially $\pi(0) = e_k$, with generator

$$Q_{j-1,j}(t) = a_j(t), \qquad Q_{j,j}(t) = -a_j(t) \quad (j \geq 1), \qquad Q_{\cdot,0} = 0.$$

The state equation is $\pi' = Q\pi$. Define

$$W^a(x) = \int_0^1 \sum_{j=1}^{k} \pi_j^a(t) \min\{x, a_j(t)\}\, dt.$$

An equal-mesh threshold is constant on each interval of some finite equal partition of $[0,1]$. Let $\mathcal{C}_k$ be the pointwise closure on $(0,\infty)$ of the convex hull of their functions W^a.

A profile v is a nonzero nonnegative nonincreasing step function of finite height, supported in $(0, S]$ for some finite S. Write $M_v = \int_0^S v(s)\, ds$, let v_1 be its maximum height, and put

$$\Psi_v(z) = \int_0^S (v(s) - z)^+\, ds \quad (z \in \mathbb{R}).$$

Let $V_0^v = 0$ and define V_j^v, $1 \leq j \leq k$, by

$$(V_j^v)'(t) = -\Psi_v(V_j^v(t) - V_{j-1}^v(t)), \qquad V_j^v(1) = 0. \tag{1}$$

Theorem 2. *For every fixed integer $k \geq 1$, the limit of $\gamma_{n,k}$ exists and*

$$\lim_{n\to\infty} \gamma_{n,k} = \inf_v \frac{V_k^v(0)}{\int v\, dP_k} = \max_{W \in \mathcal{C}_k} \inf_{x>0} \frac{W(x)}{P_k(x)}.$$

The infimum is over the profiles just defined. The maximum is attained in $\mathcal{C}_k$; no attainment by a bounded threshold policy or by a profile is asserted. This statement does not identify the limit with $\inf_n \gamma_{n,k}$ or give a rate uniform in k.

2.1 Finite-horizon lower transfer

For $n \geq k$ and $0 \leq x \leq n$, define

$$P_{n,k}(x) = \int_0^x \Pr(\mathrm{Bin}(n-1, s/n) \leq k-1)\, ds.$$

Summing the beta integrals for ranks $0, \ldots, k-1$ gives

$$P_{n,k}(n) = n \sum_{r=0}^{k-1} \binom{n-1}{r} \int_0^1 q^r (1-q)^{n-1-r}\, dq = k. \tag{2}$$

Let a have T time pieces and threshold bound A. For $n \geq \max\{k, 2A\}$, at step i use $t_i = (i-1)/n$ and accept iff $nU_i \leq a_j(t_i)$ with $j \geq 1$ slots remaining. With $p_{j,i}$ the pre-step state probabilities, set

$$W_n^a(x) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{k} p_{j,i} \min\{x, a_j(t_i)\}.$$

For a finite mixture write $\overline{W}_n$ and $\overline{W}$ for the weighted cumulative payoffs, and put

$$g_n = \inf_{0<x\leq n} \overline{W}_n(x)/P_{n,k}(x).$$

Lemma 3. *Every such mixture guarantees* $\mathrm{DP}_{n,k}(F) \geq g_n \mathrm{PR}_{n,k}(F)$ *for every admissible* F*, and*

$$\liminf_{n\to\infty} g_n \geq c := \inf_{x>0} \frac{\overline{W}(x)}{P_k(x)}.$$

For a single policy, writing

$$D = e(4A^2 + 4AT) + 2A + 2T, \qquad \delta_n = D/n,$$

one has $|W_n^a(x) - W^a(x)| \leq \delta_n x$ *and*

$$|P_{n,k}(x) - P_k(x)| \leq \frac{x^3/3 + x^2/2}{n}. \tag{3}$$

For a fixed finite mixture use the maxima of its component bounds.

Proof. Let $\psi(q) = \inf\{z : F(z) \geq 1-q\}$ and realize $X_i = \psi(U_i)$ with independent uniform U_i. Observed atoms are handled by drawing the conditional quantile using fresh randomness; this gives admissible randomized decisions and the stated iid joint realization. Put $v_n(s) = \psi(s/n)$. The pre-step state is independent of U_i, so the policy reward is $\int_0^n v_n\, dW_n^a$. An item at quantile q is among the top k precisely when at most $k-1$ other uniform quantiles are smaller. Thus $\mathrm{PR}_{n,k}(F) = \int_0^n v_n\, dP_{n,k}$. Ties among observed values are harmless. Every strict level set of nonincreasing v_n is an initial interval; Tonelli's layer-cake identity therefore turns $\overline{W}_n \geq g_n P_{n,k}$ into the claimed reward inequality. Finite mean makes both rewards finite. Mixtures are selected before arrivals, and the same argument applies to their weighted payoffs.

In the induced ℓ_1 matrix norm, $\|Q(t)\| \leq 2A$, independent of the number of states. The discrete kernel is $K_i = I + Q(t_i)/n$. If there is no threshold breakpoint inside its cell, its exact continuous propagator E_i satisfies

$$\|E_i - K_i\| \leq (2A/n)^2 e^{2A/n}.$$

At most $T-1$ cells contain a breakpoint; on these the bound $4Ae^{2A/n}/n$ suffices. All propagators are stochastic and hence contractive in ℓ_1. Telescoping gives

$$\max_i \|p_{\cdot,i} - \pi(t_i)\|_1 \le e(4A^2 + 4AT)/n.$$

Within a cell $\|\pi(t) - \pi(t_i)\|_1 \le 2A(t - t_i)$. The integrand entries $\min\{x, a_j\}$ are bounded by x; separating the breakpoint cells in the Riemann sum gives the displayed $\delta_n x$ bound.

For completeness, $d_{\mathrm{TV}}(\mathrm{Bern}(p), \mathrm{Pois}(p)) = p(1-e^{-p}) \le p^2$ for $0 \le p \le 1$. Coupling independent summands and then Poisson variables differing in mean by s/n gives

$$d_{\mathrm{TV}}(\mathrm{Bin}(n-1, s/n), \mathrm{Pois}(s)) \le (s^2 + s)/n.$$

Apply this to the event $\{0, \dots, k-1\}$ and integrate to obtain (3). Mixture errors are convex combinations of the component errors.

If $c = 0$, nonnegativity of g_n proves the conclusion directly, including the case $A = 0$. Suppose $c > 0$, so $A > 0$, and set $\kappa = P_k(A)/A > 0$. For sufficiently large n, $c - \delta_n/\kappa > 0$. Concavity gives $P_k(x) \ge \kappa x$ on $(0, A]$, and hence

$$\frac{\overline{W}_n(x)}{P_{n,k}(x)} \ge \frac{c - \delta_n/\kappa}{1 + (A^2/3 + A/2)/(\kappa n)}.$$

The positive numerator justifies replacing the denominator by its upper bound. For $A \le x \le n$, the numerator is constant and $P_{n,k}(x) \le k$. Since $\overline{W}(A) = \overline{W}(\infty) \ge kc$,

$$\frac{\overline{W}_n(x)}{P_{n,k}(x)} \ge c - \delta_n A/k.$$

Both lower bounds tend to c. □

2.2 Verification and quantitative upper transfer

Lemma 4. *The system* (1) *has a unique solution and* $V_j \ge V_{j-1} \ge 0$ *for all* $1 \le j \le k$. *For uniformly bounded Borel acceptance sets* $\mathcal{A}_j(t)$ *with jointly Borel sets* $\{(t, s) : s \in \mathcal{A}_j(t)\}$, *the associated continuum policy reward is at most* $V_k(0)$. *Equality holds for*

$$\mathcal{A}_j^*(t) = \{s : v(s) > V_j(t) - V_{j-1}(t)\}.$$

If $\Phi_0 = 0$ *and continuous piecewise absolutely continuous functions* Φ_j *satisfy* $\Phi_j(1) \ge 0$ *and* $\Phi_j' \le -\Psi_v(\Phi_j - \Phi_{j-1})$ *almost everywhere, then* $V_j \le \Phi_j$ *for every* j.

Proof. The function Ψ_v is nonnegative, nonincreasing and globally S-Lipschitz on $\mathbb{R}$, so the finite-dimensional vector field is globally Lipschitz and the terminal solution exists uniquely. First $V_1 \ge 0$. Inductively let $D_j = V_j - V_{j-1}$ and suppose $D_{j-1} \ge 0$. On an interval where $D_j < 0$,

$$D_j' = \Psi_v(D_{j-1}) - \Psi_v(D_j) \le 0.$$

Such a negative interval cannot return to the terminal value $D_j(1) = 0$, proving $D_j \ge 0$.

Write $b_j = |\mathcal{A}_j|$, $r_j = \int_{\mathcal{A}_j} v$, and $\Delta_j = V_j - V_{j-1} \ge 0$. Then $\Psi_v(\Delta_j) \ge r_j - b_j\Delta_j$, even if acceptance extends beyond the support of v. For the corresponding state law,

$$\frac{d}{dt}\sum_{j=0}^{k} \pi_j V_j = \sum_{j=1}^{k} \pi_j[-\Psi_v(\Delta_j) - b_j\Delta_j] \le -\sum_{j=1}^{k} \pi_j r_j.$$

Integration from 0 to 1 gives $J \le V_k(0)$. For the displayed maximizing sets, $\Psi_v(\Delta_j) = r_j - b_j\Delta_j$, giving equality. These sets are bounded by S, Borel, and initial intervals up to irrelevant endpoints, hence are bounded measurable quantile thresholds.

For the final assertion, proceed by induction in j. If $V_{j-1} \le \Phi_{j-1}$ then $V_j' \ge -\Psi_v(V_j - \Phi_{j-1})$. On a positive component of $V_j - \Phi_j$, its derivative is nonnegative by monotonicity of Ψ_v, so it cannot return to its nonpositive terminal value. Thus that positive component cannot exist. □

Proposition 5. *Let $v = v_m$ on $(e_{m-1}, e_m]$, with $e_M = S$ and $v = 0$ beyond S. For $n \ge \max\{k, 2S\}$, let F_n put mass $(e_m - e_{m-1})/n$ at v_m and the remaining mass at zero. Then*

$$\mathrm{DP}_{n,k}(F_n) \le V_k(0) + \frac{(2eS^2 + S)M_v}{n},$$

$$\mathrm{PR}_{n,k}(F_n) \ge \int v\, dP_k - \frac{v_1(S^3/3 + S^2/2)}{n}.$$

Consequently $\limsup_n \gamma_{n,k} \le V_k(0)/\int v\, dP_k$.

Proof. Backward induction gives a deterministic optimal finite-horizon policy whose decisions depend on time, remaining slots and current value. It may reject zero without loss. In quantile coordinates its acceptance sets $\mathcal{A}_{i,j}$ lie in $(0, S]$. Put $\alpha_{i,j} = \int_{\mathcal{A}_{i,j}} v \le M_v$; its reward is $n^{-1}\sum_{i,j} p_{j,i}\alpha_{i,j}$.

Use the same sets constantly on the corresponding continuum cells. Their generators have norm at most $2S$, so

$$\|E_i - K_i\| \le 2eS^2/n^2, \quad \max_i \|p_{\cdot,i} - \pi(t_i)\|_1 \le 2eS^2/n, \quad \|\pi(t) - \pi(t_i)\|_1 \le 2S(t - t_i).$$

Multiplying by the reward-vector bound M_v and summing cells gives

$$n^{-1}\sum_{i,j} p_{j,i}\alpha_{i,j} \le \int_0^1 \sum_j \pi_j(t)\alpha_j(t)\, dt + (2eS^2 + S)M_v/n.$$

The integral is at most $V_k(0)$ by Lemma 4. The prophet identity and layer cake, using (3) on $[0, S]$, give the denominator bound. Its limit is positive because v is nonzero. Divide for sufficiently large n and use $\gamma_{n,k} \le \mathrm{DP}_{n,k}(F_n)/\mathrm{PR}_{n,k}(F_n)$. □

For numerical certification at any fixed k, the verification lemma permits a conservative backward grid construction, computed in increasing state order at each time step:

$$\Phi_j(t_i) \ge \Phi_j(t_{i+1}) + h\Psi_v\big(\Phi_j(t_{i+1}) - \Phi_{j-1}(t_i)\big), \qquad \Phi_j(1) = 0,$$

with upward rounding. Increments are nonnegative. Linear interpolation satisfies $\Phi_j(t) - \Phi_{j-1}(t) \ge \Phi_j(t_{i+1}) - \Phi_{j-1}(t_i)$, which proves the required slope inequality. This is a valid general scheme, not a new numerical witness.

2.3 Compact minimax and control of both boundaries

Proof of Theorem 2. Let c denote the infimum over profiles in the theorem. The upper transfer gives $\limsup_n \gamma_{n,k} \le c$. Clearly $c \ge 0$; also $c \le 1$, since for $v = \mathbf{1}_{(0,x]}$ both the optimal reward and prophet reward are $P_k(x)$, obtained by taking up to k positive Poisson arrivals.

Let $\mathcal{K}_k$ be the set of nondecreasing concave functions $W : (0, \infty) \to [0, k]$ with $W(x) \le x$. Every W^a lies in it: its slope is at most $\int_0^1 \sum_{j \ge 1} \pi_j \le 1$, and its limit at infinity is the expected number

of acceptances, at most k. The defining inequalities of $\mathcal{K}_k$ are closed finite-evaluation conditions. Thus it is a convex closed subset of the compact product $[0,k]^{(0,\infty)}$, and $\mathcal{C}_k$ is compact and convex. Every member extends by $W(0)=0$ and is 1-Lipschitz: for $0<x<y$, concavity gives

$$0 \le \frac{W(y)-W(x)}{y-x} \le \frac{W(x)}{x} \le 1.$$

For each profile, the optimal bounded measurable thresholds from Lemma 4 can be approximated in L^1 by equal-mesh thresholds with values in $[0,S]$. Indeed the cell averages converge in L^1. For $\|a-b\|_{L^1} = \int_0^1 \sum_{j=1}^k |a_j - b_j|$, stochastic contraction and Duhamel's formula give

$$\sup_t \|\pi^a(t) - \pi^b(t)\|_1 \le 2\|a-b\|_{L^1}.$$

The reward integrals differ by at most $v_1|a_j-b_j|$ in state j, so

$$|J_v(a) - J_v(b)| \le (2M_v + v_1)\|a-b\|_{L^1}.$$

Consequently the supremum of the profile reward over equal-mesh policies is $V_k^v(0)$.

Write profiles as $v_\nu(s) = \nu((s,\infty))$, where ν is a nonzero finitely supported positive measure on $(0,\infty)$. Fubini gives $J_{v_\nu}(a) = \int W^a \, d\nu$ and $\int v_\nu \, dP_k = \int P_k \, d\nu$. Values at profile endpoints affect none of these expressions or the ODE. Homogeneity of (1) permits normalization $\int P_k \, d\nu = 1$; let $\mathcal{N}_k$ be this convex set of normalized finite measures. Finite evaluation makes $W \mapsto \int W \, d\nu$ continuous and affine. It has the same supremum over the generating policies as over their convex hull and closure, so compactness gives

$$\max_{W\in\mathcal{C}_k} \int W \, d\nu = V_k^{v_\nu}(0), \qquad c = \inf_{\nu\in\mathcal{N}_k} V_k^{v_\nu}(0).$$

We give the compact minimax argument explicitly. For each $x>0$ set $H_x = \{W \in \mathcal{C}_k : W(x) \ge cP_k(x)\}$. If a finite subfamily indexed by $x_1,\dots,x_m$ had empty intersection, the compact convex set of vectors

$$\left\{ \left(\frac{W(x_\ell)}{P_k(x_\ell)} \right)_{\ell=1}^m : W \in \mathcal{C}_k \right\}$$

would be disjoint from $[c,\infty)^m$. Strict finite-dimensional separation supplies nonnegative weights λ_ℓ with $\sum_\ell \lambda_\ell = 1$ such that

$$\max_{W\in\mathcal{C}_k} \sum_\ell \lambda_\ell \frac{W(x_\ell)}{P_k(x_\ell)} < c.$$

The separating weights are nonnegative because the upper orthant is unbounded in each positive coordinate direction. But $\nu = \sum_\ell \lambda_\ell \delta_{x_\ell}/P_k(x_\ell)$ lies in $\mathcal{N}_k$, contradicting the definition of c and the preceding payoff equality. The closed sets H_x therefore have the finite-intersection property, and compactness supplies $W^* \in \mathcal{C}_k$ with $W^* \ge cP_k$ everywhere. Conversely, for every $W \in \mathcal{C}_k$ and $\nu \in \mathcal{N}_k$,

$$\inf_{x>0} \frac{W(x)}{P_k(x)} \le \int W \, d\nu \le V_k^{v_\nu}(0).$$

Taking the infimum over ν proves the minimax equality and attainment in $\mathcal{C}_k$ without requiring $\mathcal{N}_k$ compact.

If $c=0$, nonnegativity and the upper transfer already prove limit existence. Suppose $c>0$ and choose $0<\varepsilon<\min\{c,1/2\}$. Choose $0<\delta<1<L$ with

$$P_k(\delta)/\delta \ge 1-\varepsilon, \qquad P_k(L) \ge k(1-\varepsilon).$$

Pointwise closure and the common 1-Lipschitz bound give a finite convex combination W of equal-mesh policy payoffs with

$$\sup_{x\in[\delta,L]} |W(x) - W^*(x)| \le \varepsilon P_k(\delta).$$

To see this directly, approximate at finitely many sufficiently fine grid points, then apply the Lipschitz bounds between them. Thus $W/P_k \ge c - \varepsilon$ on $[\delta, L]$. At the lower boundary, concavity and $P_k(x) \le x$ give, for $0 < x < \delta$,

$$\frac{W(x)}{P_k(x)} \ge \frac{W(\delta)}{\delta} \ge (c-\varepsilon)(1-\varepsilon).$$

At the upper boundary, monotonicity and $P_k(x) \le k$ give, for $x > L$,

$$\frac{W(x)}{P_k(x)} \ge \frac{W(L)}{k} \ge (c-\varepsilon)\frac{P_k(L)}{k} \ge (c-\varepsilon)(1-\varepsilon).$$

The fixed finite mixture representing W is eligible for Lemma 3; hence

$$\liminf_n \gamma_{n,k} \ge (c-\varepsilon)(1-\varepsilon).$$

Let $\varepsilon \downarrow 0$ and combine with the upper transfer. The order of choices is k, then ε, then the finite mixture, then sufficiently large n. No joint uniform convergence over all distributions or policies has been assumed. □

2.4 Occupancy-weighted domination of finite mixtures

The following constructive consequence is naturally expressed in the state occupancy and acceptance flow framework of Jiang, Ma and Zhang [9]. We do not claim a novel purification principle. It clarifies the role of mixtures as computational representations.

Proposition 6. *Every finite mixture of bounded measurable quantile policies is dominated, simultaneously for all $x > 0$, by one bounded measurable time- and state-dependent quantile policy. The corresponding statement holds at each finite horizon for feasible quantile thresholds.*

Proof. Let the mixture weights be μ_r, thresholds $a_{r,j}$ and state probabilities $\pi_{r,j}$. Put

$$p_j(t) = \sum_r \mu_r \pi_{r,j}(t), \qquad b_j(t) = \begin{cases} \dfrac{\sum_r \mu_r \pi_{r,j}(t) a_{r,j}(t)}{p_j(t)}, & p_j(t) > 0, \\ 0, & p_j(t) = 0. \end{cases}$$

These are measurable thresholds bounded by the maximum component threshold. If $p_j = 0$ all the nonnegative weighted occupancy terms vanish, so $p_j b_j = \sum_r \mu_r \pi_{r,j} a_{r,j}$ in all cases. Writing $p_{k+1} b_{k+1} = 0$ and $b_0 = 0$, the averaged state equations are

$$p_j' = p_{j+1} b_{j+1} - p_j b_j \quad (0 \le j \le k), \qquad p(0) = e_k.$$

Uniqueness for this bounded measurable linear system identifies p with the state probabilities of policy b. Concavity of $u \mapsto \min\{x, u\}$ gives

$$p_j(t) \min\{x, b_j(t)\} \ge \sum_r \mu_r \pi_{r,j}(t) \min\{x, a_{r,j}(t)\}.$$

Summing and integrating proves $W^b \ge \sum_r \mu_r W^{a_r}$. At a finite horizon define $b_{j,i}$ by the same weighted formula using pre-step state probabilities. The acceptance flows agree, so the state probabilities agree by induction. The same concavity inequality proves domination of W_n and hence, by layer cake, reward domination for every admissible law. □

The dominating continuum thresholds need not retain an equal mesh. The proposition implies equality of the game value with the *supremum* guarantee over bounded measurable deterministic quantile thresholds, because each approximating finite mixture can be dominated. Conversely, every bounded measurable threshold payoff belongs to $\mathcal{C}_k$: approximate its thresholds in L^1 by equal-mesh cell averages and apply the reward-continuity estimate above to each indicator profile $v = \mathbf{1}_{(0,x]}$. This gives pointwise convergence of the cumulative payoffs. The maximum over $\mathcal{C}_k$ therefore also bounds the deterministic supremum from above. This does not prove attainment of that supremum, nor remove the separate quantile randomization needed for atoms in F. The analytic results of this section provide no numerical bracket or nonlinear-system separation for $k \geq 3$; such conclusions require their own witnesses and certificates.

3 Certified two-unit bounds

In this section $k = 2$, $P = P_2 = 2 - e^{-x}(2 + x)$, and $V_j = V_j^v$. Thus $P''(x) = -xe^{-x} \leq 0$. Write $\bar{W}$ for the weighted cumulative acceptance function of a finite mixture. The next two propositions certify particular rational inputs; they do not certify the search procedure or its optimality.

Proposition 7 (computer-assisted)**.** *For the explicit mixture of* 36 *rational quantile policies with* $T = 64$ *in the data file* `mix_fine.json` *(rational weights summing to* 1*),* $\bar{W}(x) \geq 0.83695\,P(x)$ *for all* $x \geq 0$*.*

Proof (certificate). On each time piece the chain of each policy has a constant generator M; both π at the end of the piece and $\int \pi\,dt$ over the piece are read off from $\exp\left(h \begin{smallmatrix} M & I \\ 0 & 0 \end{smallmatrix}\right)$, $h = 1/64$, computed as a rigorous ball. This gives $\bar{W}(x) = \sum_b w_b \min\{x, b\}$ over the 1319 distinct positive thresholds b, with certified weights w_b. So $\bar{W}$ is piecewise linear and $g = \bar{W} - 0.83695P$ is convex on every linear piece.

- First piece: $g(0) = 0$ and $g'(0) = \sum_b w_b - 0.83695 > 4.04 \times 10^{-6} > 0$.
- Tail: beyond the largest threshold, $g \geq \bar{W}(\infty) - 2(0.83695) > 2.57 \times 10^{-5} > 0$, since $P < 2$.
- Each remaining piece is split into 40 subintervals, on which the tangent lines of g at the endpoints give valid lower bounds. The smallest certified bound exceeds 2.95×10^{-8}, on $[0.00558303, 0.00751685]$ (exact decimal endpoints).

All computations use python-flint 0.9.0 (Arb) at 256 bits [10]. □

Proposition 8 (computer-assisted)**.** *For the explicit rational measure with* 41 *layer atoms* ν *of the data file* `inst_do450.json`*, the profile* $v = \nu((\cdot, \infty))$ *satisfies* $V_2(0)/\int v\,dP \leq 0.8372914$*.*

Proof (certificate). On the grid $t_i = i/N$, $N = 80000$, put $\Phi_j(1) = 0$ and, backwards,

$$\Phi_1(t_i) = \Phi_1(t_{i+1}) + N^{-1}\Psi\big(\Phi_1(t_{i+1})\big), \qquad \Phi_2(t_i) = \Phi_2(t_{i+1}) + N^{-1}\Psi\big(\Phi_2(t_{i+1}) - \Phi_1(t_i)\big),$$

rounded up, and interpolate linearly. Both are nonincreasing. On $[t_i, t_{i+1}]$ we have $\Phi_1 \geq \Phi_1(t_{i+1})$ and $\Phi_2 - \Phi_1 \geq \Phi_2(t_{i+1}) - \Phi_1(t_i)$, so, Ψ being nonincreasing, the slope conditions of Lemma 4 hold. The benchmark is $\int v\,dP = \sum_j \nu_j P(x_j)$. In ball arithmetic at 200 bits, $\Phi_2(0) \leq 0.8372913936$ and $\int v\,dP \geq 0.9999999999$. □

4 An upper bound for the nonlinear-system value

The system of [1] reads, on $[0,1)$ with $y_j(0)=1$ and $y_j(t)\to 0$ as $t\uparrow 1$, $\Gamma_\ell(x)=\int_x^\infty s^{\ell-1}e^{-s}\,ds$,

$$\big(\Gamma_2(-\ln y_2)\big)' = 2-\tfrac{1}{\theta_2}-\Gamma_3(-\ln y_2), \qquad \big(\Gamma_2(-\ln y_1)\big)' = 2-\Gamma_3(-\ln y_1)-\tfrac{\theta_2}{\theta_1}\big(2-\Gamma_3(-\ln y_2)\big).$$

Write $u_j=-\ln y_j$ and $z_j=\Gamma_2(u_j)=(1+u_j)e^{-u_j}\in(0,1]$. Also put $\Gamma_3(u)=(u^2+2u+2)e^{-u}$ and $H(z)=2-\Gamma_3(\Gamma_2^{-1}(z))$. Since $dH/dz=-\Gamma_2^{-1}(z)\le 0$, H is nonincreasing, and the system reads $z_2'=H(z_2)-1/\theta_2$, $z_1'=H(z_1)-r\,f(t)$ with $r=\theta_2/\theta_1$ and $f=2-\Gamma_3(u_2)\in[0,2]$.

Lemma 9. *Any solution has $\theta_2<1/2$ and $\mathcal{T}(\theta_2)=1$, where $\mathcal{T}(\theta)=\int_0^\infty ue^{-u}\,du/(1/\theta-2+\Gamma_3(u))$. The function $\mathcal{T}$ is increasing, and $\mathcal{T}(0.48300)\le 0.99990<1<1.00009\le\mathcal{T}(0.48302)$, so $\theta_2^\star\in(0.48300,0.48302)$.*

Proof. If $\theta_2\ge 1/2$, then $z_2'\ge-\Gamma_3(u_2)$. Since $\Gamma_3(u)=(1+u)\Gamma_2(u)+e^{-u}\le(2+u)\Gamma_2(u)$, and $u_2\le 2\ln(1/z_2)$ once $u_2\ge 3$, this gives $z_2'\ge -Cz_2\ln(1/z_2)$ near 0, so z_2 cannot reach 0 in finite time; the integral also diverges at $\theta=1/2$. For $\theta_2<1/2$, z_2 is strictly decreasing and its hitting time of 0 is $\mathcal{T}(\theta_2)=1$. The enclosures use rigorous Arb quadrature on $[0,60]$ with the tail over $[60,\infty)$ bounded by $61e^{-60}/(1/\theta-2)$. □

Proposition 10 (computer-assisted)**.** *For any solution, $r=\theta_2/\theta_1>1.37$. Consequently $\theta_1^\star+\theta_2^\star<0.48302\,(1+1/1.37)<0.835590$.*

Proof. Since H is nonincreasing, a continuous piecewise-linear ϕ with $\phi(0)\le z(0)$ and $\phi'\le H(\phi)-K(t)$ almost everywhere is a lower barrier for any solution of $z'=H(z)-\tilde K(t)$ with $\tilde K\le K$: where $\phi>z>0$, $(\phi-z)'\le H(\phi)-H(z)\le 0$. Both barriers start at $\phi_j(0)=1=z_j(0)$ and live on a grid of step $1/2000$. While $\phi_2(t_i)$ is certified positive, ϕ_2 has slope $\min\{0,\underline{H}(\phi_2(t_i))-K\}$ on $[t_i,t_{i+1}]$, with $K=1/0.48300\ge 1/\theta_2^\star$ and a certified lower bound $\underline{H}$ evaluated only at certified points of $(0,1]$; once ϕ_2 is not certified positive the construction stops, since $z_2\ge 0$ holds trivially (the certificate records this with the sentinel value -1). This gives $z_2\ge\phi_2^+$ on $[0,1)$. Because z_2 is decreasing and $f=2-\Gamma_3(u_2)$ is a decreasing function of z_2, $f\le\bar f_i:=2-\Gamma_3(\overline{u}(\phi_2(t_{i+1})))$ on $[t_i,t_{i+1}]$ while $\phi_2(t_{i+1})$ is certified positive, where $\overline{u}(z)\ge\Gamma_2^{-1}(z)$ is a certified rational upper bound, and $\bar f_i=2$ afterwards. If $r\le 1.37$, the barrier ϕ_1 built with $K=1.37\bar f_i$ satisfies $z_1\ge\phi_1\ge 0.0105>0$ on $[0,1)$, contradicting $z_1(t)\to 0$. Finally $\theta_1^\star+\theta_2^\star=\theta_2^\star(1+1/r^\star)$. □

Proof of Theorem 1. Existence follows from Theorem 2 with $k=2$. Proposition 7 and Lemma 3 give the lower bound. Proposition 8 and Proposition 5 give the upper bound, since $0.8372914<0.83730$. Proposition 10 gives $\theta_1^\star+\theta_2^\star<0.48302(1+1/1.37)=5723787/6850000$. Finally, exact rational subtraction gives

$$\frac{83695}{100000}-\frac{5723787}{6850000}=\frac{18641}{13700000}>0.$$

□

5 Discussion and scope

What the separation establishes. The cumulative formulation permits a certified construction above the nonlinear-system value. It does not determine why that particular value falls short. The loss may lie in the dual restriction used in [1], in the feasible construction chosen within that restriction, or in both. In particular, we do not establish suboptimality of the entire restricted pointwise programme or necessity of a nonzero dual variable η. The finite-horizon linear-programming characterisation and the lower-bound theorem of [1] remain unaffected.

Computation versus proof. A floating-point double-oracle search used a 320-point quantile grid, 64 time pieces and 450 iterations. The reported grid value 0.837050 and profile estimate 0.837281 are discovery diagnostics, not certified continuum bounds. Restricting the adversary to a grid can overstate a policy's off-grid guarantee. Only the explicit ball-arithmetic inequalities in Propositions 7, 8 and 10, combined with the analytic transfers, justify Theorem 1. No numerical interval for $k \geq 3$ is established here.

Mixtures are convenient finite representations for search and certification. Proposition 6 shows that a bounded measurable deterministic quantile policy can dominate each finite mixture. Therefore the better certificate from the 36-policy mixture than from the illustrative 24-piece policy in Appendix A is not evidence that mixing is intrinsically superior. The recovered threshold need not retain the original time mesh; attainment of the optimal value by one bounded threshold policy remains unproved.

Remaining questions. Theorem 2 concerns each fixed k, not a budget growing with n. Its maximum is attained in a compact closure, not necessarily by a single implementable optimizer. It proves neither equality of the limit with the all-horizon infimum nor an exact numerical constant. Quantitative transfer bounds depend on the chosen profile or policy, and are not a uniform convergence rate over the entire continuum game. Sharper certificates for larger budgets and a characterization of extremizers are separate problems.

Reproducibility and provenance. The accompanying repository contains the rational policy mixture, layer measure, illustrative policy, certificate programs and replay documentation. Threshold-array row j, counted from 1, contains the threshold for j remaining slots. A layer atom (x, w) specifies a jump of height w in the profile; it is not an atom of an observation distribution. The upper transfer specifies the actual horizon-dependent observation law.

The analytic proof covers every fixed budget. Numerical replay covers only the supplied finite $k = 2$ witnesses and the implemented arithmetic checks; it is not a formal verification of the manuscript, a proof of global numerical optimality, or an independent human peer review. The revision separates historical search outputs from certificate evidence and does not treat unsupported older numerical upper estimates as results. Automated research assistance was used in proof development, revision and internal checking. The supplied review is advisory evidence, not an external publication endorsement.

Declarations. The study uses mathematical constructions and no human-participant or personal data; ethics approval is not applicable. Data and code are prepared for the public repository `https://github.com/ipitchford/iid-multiunit-prophet-limits`, with the versioned archive assigned the reserved DOI above. These identifiers do not by themselves assert that public release has completed. Authorship is anonymous. No claim about undisclosed funding or competing interests is made.

A An illustrative two-unit threshold policy

Proposition 11 (computer-assisted)**.** *For the rational deterministic policy of Table 1 ($T = 24$), $W(x) \geq 0.8365\, P(x)$ for all $x \geq 0$.*

Certificate. As for Proposition 7 with a single policy: $g'(0) > 1.85 \times 10^{-5} > 0$, the tail margin exceeds 0.0028, and the smallest certified lower bound over the 47 remaining pieces (400 subintervals

each) exceeds 3.2×10^{-5}. These are conservative lower bounds, not equalities obtained by rounding displayed decimals. □

Table 1: The certified policy ($k = 2$, $T = 24$). On each time piece the policy accepts an arrival with scaled tail quantile $s \le a_2$ when two slots remain, and $s \le a_1$ when one remains. All values are exact decimals.

time piece	a_2 (two slots)	a_1 (one slot)
$[0/24, 1/24)$	1.170	0.184
$[1/24, 2/24)$	1.303	0.123
$[2/24, 3/24)$	1.433	0.125
$[3/24, 4/24)$	1.643	0.158
$[4/24, 5/24)$	1.781	0.252
$[5/24, 6/24)$	1.911	0.306
$[6/24, 7/24)$	2.130	0.370
$[7/24, 8/24)$	2.272	0.439
$[8/24, 9/24)$	2.413	0.507
$[9/24, 10/24)$	2.650	0.581
$[10/24, 11/24)$	2.801	0.652
$[11/24, 12/24)$	2.952	0.733
$[12/24, 13/24)$	3.224	0.809
$[13/24, 14/24)$	3.374	0.889
$[14/24, 15/24)$	3.662	0.977
$[15/24, 16/24)$	3.883	1.072
$[16/24, 17/24)$	4.086	1.547
$[17/24, 18/24)$	4.483	2.040
$[18/24, 19/24)$	4.723	2.540
$[19/24, 20/24)$	5.297	3.068
$[20/24, 21/24)$	5.620	3.593
$[21/24, 22/24)$	6.009	4.234
$[22/24, 23/24)$	8.273	5.010
$[23/24, 24/24)$	10.602	6.587

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